Zejun Bai

AI / LLM Engineer

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SUMMARY

AI engineer focused on agent workflows, multimodal retrieval, and language-model alignment. Project experience connects Python model pipelines with APIs, browser automation, and application interfaces.

TECHNICAL SKILLS

AI agents, Evaluation, FAISS, LangGraph, LLM applications, Python, RAG, Automation, Docker, FastAPI, Playwright, PostgreSQL, PPO, PyTorch, Redis, Reward modeling

SELECTED PROJECT EXPERIENCE

Zillusion

- Built an agentic data platform connecting natural-language requirements, source ranking, browser exploration, and generated collection workflows using FastAPI, LangGraph, and Playwright.
- Integrated separate workflow validation, PostgreSQL storage, Redis-backed tasks, and a Next.js workbench with streamed execution events.

Multimodal literature RAG

- Combined FAISS dense retrieval, SPLADE, BM25, and reciprocal rank fusion, using Qwen2-VL descriptions to make tables and figures searchable.
- Implemented a multimodal PDF retrieval pipeline that parses text, equations, tables, and figures into semantic chunks with document and page metadata.

RLHF pipeline

- Implemented a distilGPT-2 alignment pipeline covering supervised fine-tuning, pairwise reward-model training, and custom sequence-level PPO with KL regularization.
- Compared base and aligned model variants with reward analysis, out-of-distribution evaluation, and reward-hacking checks.

AI outreach and social operations

- Extended a Django and Playwright outreach system with an SMTP/IMAP email channel, personalized connection notes, campaign controls, and an operator console.
- Integrated OpenClaw-based Xiaohongshu operations with container orchestration, health supervision, content generation, and scheduled publishing; explored Rednote search automation.

EDUCATION

Duke University | Master of Engineering, Artificial Intelligence | Awarded May 2026

Sichuan University | Bachelor's degree in Automation | Sep 2019 - Jun 2024

National Taiwan University of Science and Technology | Electrical Engineering exchange | Sep 2023 - Jan 2024